

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

*I Krisha Meenatchi M (192511338)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Krisha Meenatchi M

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.
2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.
3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.
4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

		and coherence			
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandabl e with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Question 1

Given

- File size = 150 MB = 150,000,000 bytes
- Bandwidth = 50 Mbps
- Segment size (Transport Layer) = 1500 bytes
- DLL overhead per frame = 40 bytes (header + trailer)

Calculation

Total segments: $150,000,000 \div 1500 = 100,000$ segments

Total frames: Each segment $\rightarrow$ one frame $\Rightarrow 100,000$ frames

Frame size: $1500 + 40 = 1540$ bytes

Total DLL overhead: $100,000 \times 40 = 4,000,000$ bytes = 4 MB

Total data transmitted: $100,000 \times 1540 = 154,000,000$ bytes = 1,232,000,000 bits

Transmission time: $1,232,000,000 \text{ bits} \div 50,000,000 \text{ bps} = 24.64$ seconds

Explanation – Reliability

The Transport Layer ensures reliability by breaking the file into numbered segments, using sequence numbers, checksums, and acknowledgements (as in TCP) to detect loss or corruption, and retransmitting missing segments. It also performs flow and congestion control so the sender does not overwhelm the receiver.

The Data Link Layer ensures reliability on each physical link by framing the data with header and trailer fields containing addressing and a CRC (cyclic redundancy check). If a frame is found corrupted at the receiving end, it is discarded and, depending on the protocol, retransmission is requested, guaranteeing error-free delivery hop-by-hop.

Question 2

Given

- Window size = 6 frames
- Frame size = 1024 bytes (user data)
- Total frames sent = 48

Calculation

Number of windows: $48 \div 6 = 8$ windows

Frames lost: 1 frame lost per window $\times 8$ windows = 8 frames lost

Total retransmissions: 8 retransmissions

Explanation – Flow Control

Flow control using a sliding window allows the sender to transmit several frames (here, 6) before waiting for an acknowledgement, instead of sending one frame and waiting each time (stop-and-wait). This keeps the transmission link continuously utilized, increases throughput, and matches the sender's rate to the receiver's buffer capacity so the receiver is never overwhelmed. When a frame is lost, only the frames within that window need to be retransmitted, which keeps recovery efficient compared to retransmitting the entire message.

Question 3

Given

- Original packet size = 12,000 bytes (includes original 20-byte IP header)
- MTU = 1500 bytes
- IP header per fragment = 20 bytes

Calculation

Payload to fragment: $12,000 - 20 = 11,980$ bytes

Usable data per fragment: $1500 - 20 = 1480$ bytes

Total fragments: $\lceil 11,980 \div 1480 \rceil = \lceil 8.09 \rceil = 9$ fragments

Total header overhead: $9 \times 20 = 180$ bytes

Total bytes transmitted: $11,980 + 180 = 12,160$ bytes

Explanation – Network Layer Delivery

The Network Layer fragments packets larger than a link's MTU so they can be transmitted across that link. Each fragment carries the same Identification field (to show which original packet it belongs to), a Fragment Offset field (to show its position within the original data), and a More Fragments (MF) flag (cleared only on the last fragment). At the destination, the Network Layer uses these fields to reassemble the fragments in the correct order into the original packet before passing it up to the Transport Layer, ensuring complete and correct delivery even when fragmentation occurs along the path.

Question 4

Answer

Number of windows: $48 \div 6 = 8$ windows

Total retransmissions: 8 (one lost frame per window)

Flow control explanation: as in Question 2, the sliding window mechanism lets multiple frames be in transit before an acknowledgement is required, improving link utilization and matching sender/receiver speeds, which increases overall communication efficiency.

Question 5

Given

- Total file size = 600 MB
- Checkpoint interval = 60 MB
- Failure occurs at 390 MB transferred

Calculation

Checkpoints reached before failure: $\lfloor 390 \div 60 \rfloor = 6$ checkpoints (at 60, 120, 180, 240, 300, 360 MB)

Data to retransmit after recovery: $390 - 360 = 30$ MB

Explanation – Session Layer Synchronization

The Session Layer inserts synchronization checkpoints to manage and control the dialogue between two communicating systems over a long transfer. If a failure occurs, the session does not need to restart from zero — it resumes from the last checkpoint. This drastically reduces the amount of data that must be retransmitted (only 30 MB here instead of all 390 MB already sent), saving time, bandwidth, and reducing the impact of network failures on large data transfers such as video or file streaming.

Question 6

Given

- Original file size = 900 MB
- Compression reduces size by 40%
- Encryption adds 5% overhead to compressed size

Calculation

Compressed size: $900 \times (1 - 0.40) = 900 \times 0.60 = 540$ MB

Encrypted size: $540 \times (1 + 0.05) = 540 \times 1.05 = 567$ MB

Total % reduction vs original: $(900 - 567) \div 900 \times 100 = 37\%$

Explanation – Presentation Layer Role

The Presentation Layer improves transmission efficiency by compressing data, which reduces the volume of data that must travel across the network, lowering transmission time and bandwidth consumption. It improves security by encrypting the compressed data, protecting confidentiality and integrity in transit, even though encryption itself adds a small overhead. Even after this overhead, the file is still significantly smaller (37% smaller) than the original, showing a net efficiency gain alongside the security benefit.

Question 7

Given

- Total data = 3 GB = 3000 MB
- Request size = 3 MB
- Effective throughput = 120 Mbps

Calculation

Total requests: $3000 \div 3 = 1000$ requests

Total data in bits: $3000 \text{ MB} \times 8 = 24,000 \text{ Mb}$

Transmission time: $24,000 \text{ Mb} \div 120 \text{ Mbps} = 200$ seconds

Explanation – Application Layer Reliability

The Application Layer (FTP) provides the user-facing service for file transfer, organizing the transfer into manageable requests and handling functions such as authentication, directory navigation, and request/response sequencing. It relies on the underlying Transport Layer (typically TCP) to guarantee reliable, in-order, error-checked delivery of each request. Together, this layering allows FTP to present a simple, reliable file-transfer service to the user while the complexity of ensuring reliable delivery is handled beneath it.

Question 8

Given

- Processing delays: App = 4 ms, Transport = 3 ms, Network = 5 ms, Data Link = 2 ms, Physical = 1 ms
- Propagation delay = 18 ms
- Transmission delay = 12 ms

Calculation

Total processing delay: $4 + 3 + 5 + 2 + 1 = 15$ ms

Total end-to-end delay: $15 + 18 + 12 = 45$ ms

Explanation – Layer Contributions

The Application Layer contributes delay while preparing and formatting user data. The Transport Layer adds delay for segmentation, sequencing, and multiplexing. The Network Layer contributes routing and addressing decisions (path selection). The Data Link Layer adds framing, addressing, and error-check computation. The Physical Layer contributes the delay of converting bits to signals. In addition to these processing delays, transmission delay is the time to push all the packet's bits onto the link, and propagation delay is the time for a signal to travel across the physical medium to the receiver. All these delays add up to give the total end-to-end delay experienced by the packet.

Question 9

Given

- Useful data = 80 MB = 80,000,000 bytes
- Frame size = 1600 bytes
- Overhead per frame = 80 bytes

Calculation

Payload per frame: $1600 - 80 = 1520$ bytes

Total frames required: $\lceil 80,000,000 \div 1520 \rceil = 52,632$ frames

Total overhead transmitted: $52,632 \times 80 = 4,210,560$ bytes ≈ 4.21 MB

Transmission efficiency: $(1520 \div 1600) \times 100 = 95\%$

Explanation – Effect of Overhead

Protocol overhead is the portion of each frame used for control information (headers/trailers) rather than actual user data. As overhead increases relative to frame size, transmission efficiency drops because a larger fraction of the network's capacity is spent carrying non-data information rather than useful payload. Here, with 80 bytes of overhead in a 1600-byte frame, the network still achieves a relatively high 95% efficiency, but any increase in overhead (e.g., for extra security or routing information) would proportionally reduce the useful throughput available to the application.

Question 10

Given

- Original record size = 240 MB
- Presentation Layer compression = 30%
- Transport Layer segment size = 1400 bytes
- Data Link Layer overhead = 60 bytes per frame

Calculation

Compressed size: $240 \times (1 - 0.30) = 240 \times 0.70 = 168 \text{ MB}$

Total segments: $168,000,000 \div 1400 = 120,000 \text{ segments}$

Total DLL overhead: $120,000 \times 60 = 7,200,000 \text{ bytes} = 7.2 \text{ MB}$

Final data across Physical Layer: $168 \text{ MB} + 7.2 \text{ MB} = 175.2 \text{ MB}$

Explanation – Multi-Layer Cooperation

Each OSI layer contributes a distinct, complementary function to secure and reliable healthcare data transmission. The Presentation Layer compresses the medical record to reduce its size for efficient transmission (and can also encrypt it for confidentiality, which is critical for sensitive patient data). The Transport Layer divides the compressed data into fixed-size segments and ensures reliable, ordered, error-checked delivery end-to-end, using acknowledgements and retransmission where needed. The Data Link Layer frames each segment with addressing and error-detection overhead so that data is delivered correctly across each physical hop of the IoT-enabled network. Together, these layers ensure that the patient's medical record is transmitted compactly, securely, and reliably from source to destination.